Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY REGULAR EXAMINATIONS
2023/2024 ACADEMIC YEAR
SECOND YEAR

COURSE CODE: UCC 202

COURSE TITLE: DEVELOPMENT CONCEPTS AND APPLICATIONS

DATE: 15/04/2024 TIME: 3-5PM

INSTRUCTIONS TO CANDIDATES

Instructions: Answer question ONE and any other THREE

Time: 2 hours

QUESTION ONE (25 MARKS)

a) Distinguish between economic growth and economic development. (4 Marks)
b) Use three illustrations to explain the contradictions contained in economic dualism theory by Lewis (6mrks)
c) Explain three economic growth indictors as compared to composite indicators of economic development (6 Marks)
d) Explain the three main characteristics of endogenous development (6 Marks)
e) Outline three of dimensions of development (3 Marks)

QUESTION TWO (15 MARKS)

a) Explain five assumptions of Adam Smith theory of economic growth (10 Marks)
b) Outline five negative impacts of globalization to developing economies (5 marks)

QUESTION THREE (15 MARKS)

a) Describe four characteristics of transitional societies (8 Marks)
b) Explain two contributions of regional integration in economic growth and development in Kenya. (4 Marks)
c) Identify three benefits of participatory approach to economic development. (3 Marks)

QUESTION FOUR (15 MARKS)

a) Explain six functions of a government/state in economic development. (12 marks)
b) Highlight three weaknesses of the classical theories to developing countries. (3 Marks)

QUESTION FIVE (15 MARKS)

a) Outline five composite indicators of development (5marks)
b) Explain the stages of economic development according to W. Rostow (10 marks)

QUESTION SIX (15 MARKS)

a) Discuss five dimensions of sustainable development. (10 marks)
b) Outline five negative effects of foreign aids to developing economies (5 Marks)

Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY REGULAR EXAMINATIONS
2022/2023 ACADEMIC YEAR
SECOND YEAR

COURSE CODE: UCC 202

COURSE TITLE: DEVELOPMENT CONCEPTS AND APPLICATIONS

DATE: 11/4/2023 **TIME:9-12PM**

INSTRUCTIONS TO CANDIDATES

Instructions: Answer question **ONE** and any other **THREE**

Time: 3 hours

Please Turn Over

QUESTION ONE (25 MARKS)

a) Distinguish between economic growth and economic development. (4mrks)

b) Describe two types of economic policies (4mrks)

c) Outline three reasons why economic growth is not sufficient to eradicate poverty. (3mrks)

d) Outline four composite Indicators of Development. (4mrks)

e) Identify three benefits of participatory approach to economic development. (3mrks)

f) using an example in each case, explain two roles of technology in economic development (4mrks)

g) Identify three ways in which education may contribute to sustainable development (3mrks)

QUESTION TWO

a) Highlight four reasons why developing should be discouraged from relying on foreign aid(4mrks)

b) Identify three demerits of regional integration between Kenya and its East African neighbours. (3mrks)

c) Explain four reasons why Kenya is classified as a developing country. (8mrks)

QUESTION THREE

a) Identify three characteristics of Rostow's "stationary phase" of economic growth and development. (3mrks)

b) Explain six reasons to support that Kenyan economy may be operating at the level of traditional society phase as asserted by W.W. Rostow. (12mrks)

QUESTION FOUR

a) Explain the five objectives of the World Bank and International Monetary Fund through Structural Adjustment Programmes. (10mks)

b) Identify five negative impacts of Structural Adjustment Programmes as an approach to economic development of the Sub-Saharan Africa economies. (5mks)

QUESTION FIVE

a) Describe the meaning of the following terms and concepts as used in Economic Development

i. Common market (1mrks
ii. Globalization (1mrks)
iii. Dualism (1Mark)

b) Describe two assumptions of the Dualism theory of economic development. (4mrks)

c) Identify four limitations of dualism theory of economic development. (8mrks)

QUESTION SIX

a) Explain five roles played by the Kenyan Government in economic development (5mrks)

b) With relevant examples, explain five reasons why comparative advantage theory may not apply to Kenyan economy (10mrks)

Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY REGULAR EXAMINATIONS

2021/2022 ACADEMIC YEAR

SECOND YEAR

COURSE CODE: UCC 202

COURSE TITLE: DEVELOPMENT CONCEPTS AND APPLICATIONS

DATE: 6TH JUNE 2022 **TIME: 9 A.M-12P.M.**

INSTRUCTIONS TO CANDIDATES

Instructions: Answer question **ONE** and any other **THREE**

Time: 3 hours

This paper contains 3 pages

Please Turn Over

QUESTION ONE (25 MARKS)

a) A committee appointed by the recently elected Government in the USA has approached you to advise on the best economic policies that will enable the country realize rapid growth and development. You have advised the committee to formulate policies that allow for free trade as espoused by Adams Smith.

Required

i. Explain five reasons to support the choice (10 marks)

ii. Provide three limitations of this approach in the Kenya's context (3 marks)

b) Set out five dimensions of development (5 marks)

c) Distinguish between economic growth and economic development (4 marks)

d) Identify three ways in which education should contribute to sustainable development (3 marks)

QUESTION TWO (15 MARKS)

a) Identify five composite indicators of development (5marks)

b) Explain five reasons why it is preferable to use composite indicators when measuring economic development (10 marks)

QUESTION THREE (15 MARKS)

a) Describe four characteristics of transitional societies (8mrks)

b) Explain two contributions of regional integration in economic growth and development in Kenya (4 marks)

c) Identify three benefits of participatory approach to economic development (3 marks)

QUESTION FOUR (15 MARKS)

a) Explain five functions of a government in economic development (10 marks)

b) Provide five negative impacts of globalization to developing economies (5 marks)

QUESTION FIVE (15 MARKS)

a) Identify five limitations of GDP per capital in measuring economic growth (5marks)

b) Explain three conditions that are necessary for economic take-off stage as expounded by Rostow's theory (6 marks)

c) Explain how the vicious circle of poverty operates to perpetuate the low level of development in Africa (4 marks)

QUESTION SIX (15 MARKS)

-Social solidarity
-Economic
-Human
-Environmental

a) Discuss five dimensions of sustainable development. (10 marks)

b) Identify five negative effects of foreign aids to developing economies (5 Marks)

Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY REGULAR EXAMINATIONS
2019/2020 ACADEMIC YEAR
FIRST YEAR

COURSE CODE: UCC 202

COURSE TITLE: DEVELOPMENT CONCEPTS AND APPLICATIONS

DATE: 17TH APRIL 2020 **TIME: 2-5PM**

INSTRUCTIONS TO CANDIDATES

Instructions: Answer question **ONE** and any other **THREE**

Time: 3 hours

This paper contains 3 pages

Please Turn Over

QUESTION ONE (25 MARKS)

a) Distinguish between economic growth and economic development (4 marks)

b) Explain five reasons why it is preferable to use composite indicators when measuring economic development (10 Marks)

c) Explain three ways in which education should contribute to sustainable development (6 Marks)

d) Outline five of dimensions of development (5 Marks)

QUESTION TWO (15 MARKS)

a) Explain five assumptions of Adam Smith theory of economic growth(10 Marks)

b) Outline five negative impacts of globalization to developing economies (5 marks)

QUESTION THREE (15 MARKS)

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QUESTION FOUR (15 MARKS)

a) Explain six functions of a government in economic development (12 marks)

b) Highlight three weaknesses the classical theories to developing countries (3 Marks)

QUESTION FIVE (15 MARKS)

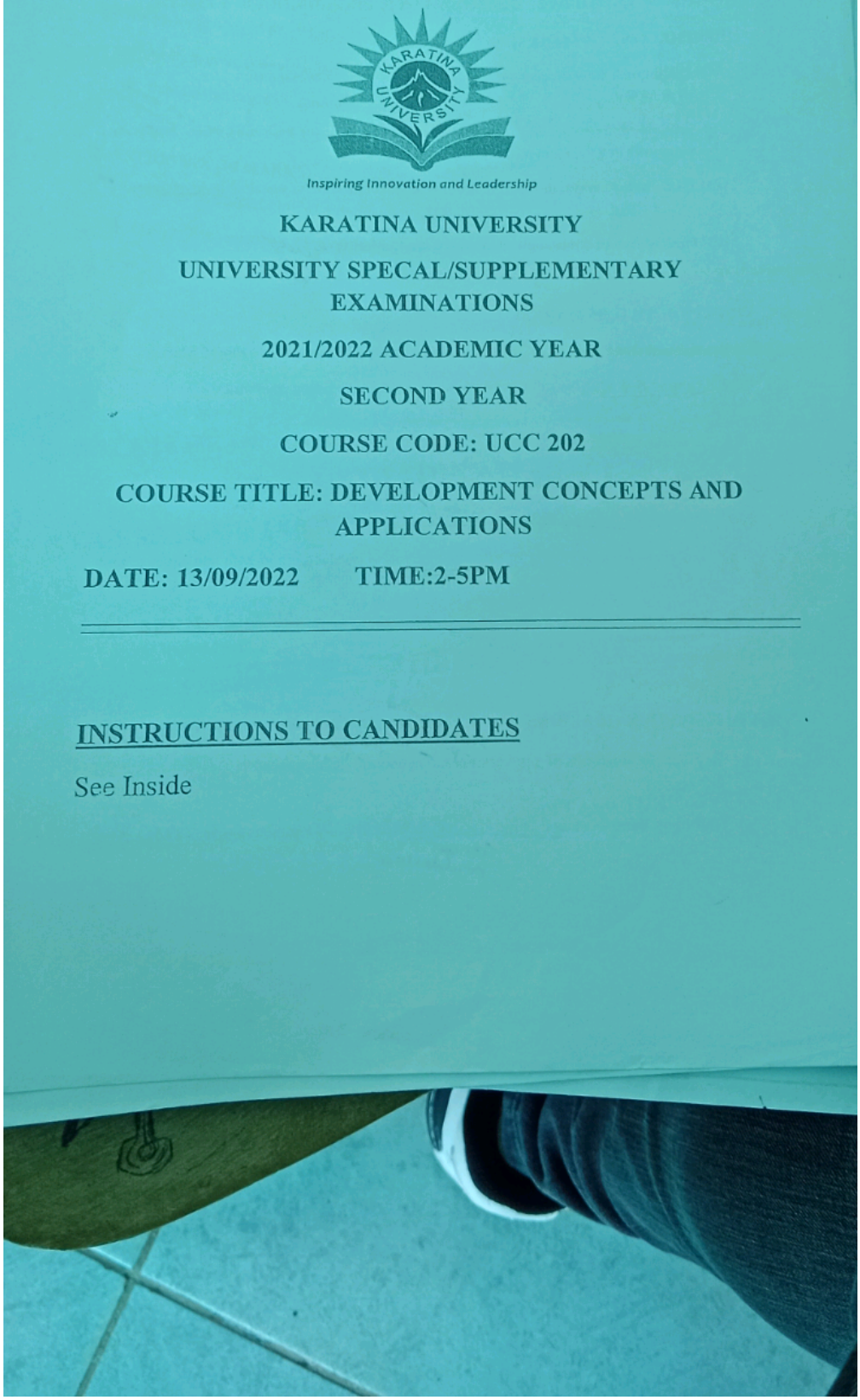
a) Outline three composite indicators of development (3marks)

b) Explain the stages of economic development according to W. Rostow (12 marks)

QUESTION SIX (15 MARKS)

a) Discuss five dimensions of sustainable development. (10 marks)

b) Outline five negative effects of foreign aids to developing economies (5 Marks)



Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY SPECAL/SUPPLEMENTARY EXAMINATIONS

2021/2022 ACADEMIC YEAR

SECOND YEAR

COURSE CODE: UCC 202

COURSE TITLE: DEVELOPMENT CONCEPTS AND APPLICATIONS

DATE: 13/09/2022 TIME:2-5PM

INSTRUCTIONS TO CANDIDATES

See Inside

Instructions: Answer question **ONE** and any other **THREE**

QUESTION ONE (25 MARKS)

a) Distinguish between economic growth and economic development (4 marks)

b) Explain five reasons why it is preferable to use composite indicators when measuring economic development (10 Marks)

c) Explain three ways in which education should contribute to sustainable development (6 Marks)

d) Outline five of dimensions of development (5 Marks)

QUESTION TWO (15 MARKS)

a) Explain five assumptions of Adam Smith theory of economic growth(10 Marks)

b) Outline five negative impacts of globalization to developing economies (5 marks)

QUESTION THREE (15 MARKS)

a) Describe four characteristics of transitional societies(8 Marks)

b) Explain two contributions of regional integration in economic growth and development in Kenya (4 Marks)

c) Identify three benefits of participatory approach to economic development (3 Marks)

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a) Explain six functions of a government in economic development (12 marks)

b) Highlight three weaknesses the classical theories to developing countries (3 Marks)

QUESTION FIVE (15 MARKS)

a) Outline three composite indicators of development (3marks)

b) Explain the stages of economic development according to W. Rostow (12 marks)

QUESTION SIX (15 MARKS)

a) Discuss five dimensions of sustainable development. (10 marks)

b) Outline five negative effects of foreign aids to developing economies (5 Marks)

COURSE CODE: UCC 202

COURSE TITLE: DEVELOPMENT CONCEPTS AND APPLICATIONS

DATE: 13/09/2022 TIME:2-5PM

INSTRUCTIONS TO CANDIDATES

See Inside

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS
2023/2024 ACADEMIC YEAR

SECOND YEAR SECOND SEMESTER EXAMINATIONS

FOR THE DEGREE OF

BACHELOR OF ARTS, BACHELOR OF SCIENCE &

BACHELOR OF EDUCATION

COURSE CODE: UCC 203

COURSE TITLE: GENDER STUDIES

DATE: APRIL 15, 2024 **TIME: 9:00-11:00AM**

Instructions to candidates

- SEE INSIDE

This paper consists of two (2) printed pages. **PLEASE TURN OVER**

Chivalry

* Impact of the Phelps Stoke commission report

KARATINA UNIVERSITY
School of Education and Social Sciences
Department of Social Sciences G 05443/24
Main & Town Campuses

Second Year Second Semester Examinations 2023/2024 Academic Year
UCC 203: GENDER STUDIES

Instructions to candidates: Answer ALL questions in SECTION A and TWO questions in SECTION B.

SECTION A (30 MARKS)

Q1(a). UN Convention on the Elimination of all Forms of Discrimination Against Women (UNCEDAW) obligates State Parties to undertake several measures to ensure gender equity. Describe these measures while explaining their applicability in Kenya (12 marks)

(b). Explain the relationship between gender and health in contemporary society (8 marks)

(c). What is the importance of studying gender at the university level? (10 marks)

SECTION B (40 MARKS)

Q2(a). Differentiate between *Women in Development* and *Gender and Development* as approaches to mainstreaming gender in development (8 marks).

(b). Explain how Role Theory may be utilized in the explanation of gendered responsibilities in society (12 marks)

Q3(a). In Kenya, women are underrepresented in senior management positions despite being more than men in the general population. Explain this anomaly (12 marks)

(b). Expound on the statement, *education and law are principal engines of gender equality in society* (8 marks)

Q4(a). bell hooks (1984) argued for the need to move women 'from margin to center'. How can this be realized in Kenya (12 marks)

(b). What bottlenecks may be encountered in 'moving women from margin to center'? (8 marks)

Q5(a). How does Social Learning Theory (Bandura, 1977) explain the statement, *Gender is a Social Construction* (12 marks)

(b). Explain the relationship between gender on one hand and power and governance on the other, while drawing apt examples from Kenya (8 marks)

Q6. Differentiate the following pairs of concepts:

(a). Marxism and Liberal Feminism (5 marks)

(b). Gender Mainstreaming and Affirmative Action (5 marks)

(c). Gender Blind and Gender Neutral Policy (5 marks)

(d). Nature and Nurture in gender differentiation (5 marks)

KARATINA UNIVERSITY
School of Education and Social Sciences
Department of Social Sciences
Main Campus
Second Year Second Semester Examinations 2022/2023 Academic Year

UCC 203: GENDER STUDIES.

Instructions: Answer all questions in SECTION A and any TWO in SECTION B

SECTION A (30 MARKS).

1 (a) Distinguish between the following terms as used in Gender Studies:
- i) Gender equity and gender equality (2 marks)
- ii) Gender roles and gender stereotypes (2 marks)
- iii) Social values and social norms (2 marks)
- iv) Patriarchy and matriarchy (2 marks)
- v) Gender analysis and gender blind (2 marks)

b) i) Outline three differences between gender and sex. (6 marks)
ii) Outline the main ways in which children learn gender behavior (4 marks)

c) Describe the following:
i) The mission statement of United Nations Sustainable Development Goal (UNSDG) number 5. (1mark)
ii) Five targets of UNSDG No. 5 . (5 marks)

d) Describe the two types of sexual harassment experienced in Kenya work places. (4 marks)

SECTION B (40 MARKS).

2. (a) Describe two agents of gender socialization in Kenya. (10 marks)
(b) Explain with aid of examples two strands of feminism. (10 marks)

3. (a) Outline four components of human development as stipulated by United Nations Development Programme (UNDP). (8 marks)
(b) Describe three gender roles development theories. (12 marks)

4. (a) Explain the concept of "Doing Gender" as applied in Gender Studies. (6 marks)
(b) Discuss two key approaches to gender and development (14 marks)

5. (a) Explain five effects of gender-based violence (GBV) to individuals and society. (10 marks)
(b) Outline five strategies of encouraging women participation in political activities in Kenya. (10 marks)

6. (a) Outline five benefits of implementing the National Gender and Development Policy in Kenya. (10 marks)
(b) Explain the following:
i) How the Kenya Constitution (2010) help to promote gender equity and equality. (6 marks)
ii) Four considerations Karatina University can put in place to guarantee gender equity and equality in the work place. (4 marks)

KARATINA UNIVERSITY
School of Education and Social Sciences
Department of Social Sciences
Main Campus

Second Year Supplementary/Special Examinations 2021/2022 Academic Year

UCC 203: GENDER STUDIES

Instructions to candidates: Answer ALL questions in SECTION A and TWO questions in SECTION B.

SECTION A (30 MARKS)

Q1(a). Give reasons for the various gender mainstreaming challenges in Kenya (10 marks)

Q1(b). How does gender-based violence manifest itself in Kenya (10 marks)

Q1(c). Gender analysis is central to women empowerment. Discuss this statement while using appropriate examples (10 marks)

SECTION B (40 MARKS)

Q2. With appropriate illustrations argue for the statement that: *women passive participation in politics is a non-starter* (20 marks).

Constitution

Q3. Research has shown that men are more exposed to violence comparative to women. Outline the various reasons behind this state of affairs (20 marks).

- Rights
- Organization
- Ar men regarded to violate
- Rights of men
- Strong

Q4. Bring out the tenets of the Gender Schema Theory and determine how it may be used to explain the development of gender roles (20 marks).

Q5. It is noted that health and illness and response take both gender and political dimensions. Explain the foregoing statement while citing relevant illustrations (20 marks).

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS FOR 2023/2024 ACADEMIC YEAR

SECOND YEAR SECOND SEMESTER EXAMINATIONS

FOR THE DEGREE OF BACHELOR OF EDUCATION (ARTS/SCIENCE/ARTS WITH EDUCATION/ GUIDANCE AND COUNSELLING/SPECIAL NEEDS EDUCATION), BACHELOR OF SCIENCE (AGRICULTURAL EXTENSION AND EDUCATION) AND BACHELOR OF SCIENCE WITH EDUCATION

COURSE CODE : CIM 211
COURSE TITLE : EDUCATIONAL MEDIA AND RESOURCES

DATE : 19/4/2024 TIME: 9AM -11 AM

INSTRUCTIONS TO CANDIDATES

SEE INSIDE

THIS PAPER CONSISTS OF TWO (2) PRINTED PAGES
PLEASE TURN OVER

Instructions to candidates

i) Answer Question One (1) (Compulsory) and any other three (3).

ii) Each new question should start on a separate page on the answer booklet

QUESTION 1 (25MAKS)

a) Giving appropriate examples from your subject area, describe five situations during instruction that a teacher would require to use a named instructional resource. (5 marks)

b) Describe five advantages of instructional film over print media. (10 marks)

c) Discuss five ways in which a teacher can make verbal communication in the classroom effective. (10marks)

QUESTION 2 (15MKS)

a) Suggest how a radio broadcast developer can make the programme interesting to the learner. (5marks)

b) Describe five criteria used when evaluating a text book as an instructional material. (10 marks)

QUESTION 3 (15 MARKS)

a) Describe how a teacher can maintain professionalism when utilizing slides during lesson presentation. (5 marks)

b) Explain five ways in which a teacher can overcome limitations of the chalkboard as an instructional medium. (10 marks)

QUESTION 4 (15 MARKS)

a Explain five reasons why a teacher uses non- verbal communication. (5 Marks)

b) Discuss five factors that a teacher should consider when using visual communication. 10 Marks

QUESTION 5 (15 MARKS)

a) Discuss any five (5) barriers to classroom communication can be overcome (5marks)

b) By means of examples, explain how the ASSURE model can guide a teacher while planning for a lesson that utilizes an instructional media. (10marks)

********END********

KARATINA UNIVERSITY
UNIVERSITY EXAMINATIONS
2023/2024 ACADEMIC YEAR
SECOND YEAR FIRST SEMESTER
REGULAR EXAMINATIONS

FOR THE DEGREE OF BACHELOR OF EDUCATION (ARTS/SCIENCE/GUIDANCE & COUNSELLING/ SPECIAL NEEDS), BACHELOR OF SCIENCE/ARTS WITH EDUCATION & BACHELOR OF SCIENCE AGRICULTURE EDUCATION AND EXTENSION

COURSE CODE: EDF 211

COURSE TITLE: HISTORY OF EDUCATION

DATE: 17/04/2024 TIME: 9AM – 11AM

INSTRUCTION TO CANDIDATES:

SEE INSIDE

THIS PAPER CONSISTS OF (2) PRINTED PAGES

PLEASE TURNOVER

INSTRUCTIONS: Answer Question ONE and any other THREE questions

QUESTION ONE (25 MARKS)

a) Explain **five** methods of teaching borrowed from Ancient Egyptian education (5 marks)

b) Discuss **five** ways through which ancient Athenian education has influenced modern education (5 marks)

c) Relate **five** forces that contributed to the rise of universities in Europe (5marks)

d) Evaluate **five** challenges facing education sector in Kenya (5 marks)

e) Justify **five** ways through which Koech education Commission Report has impacted on education in Kenya (5 marks)

QUESTION TWO (15 MARKS)

a) Explain **five** factors that hampered missionary activities in the development of education in Africa up to 1920 (5 marks)

b) Analyse **five** ways in which Islamic education has contributed to the educational and scientific development in modern times (10 marks)

QUESTION THREE (15 MARKS)

a) Discuss **five** causes of the renaissance in Europe (5 marks)

b) Examine **five** factors behind the development of education in Africa between 1920-1945 (10 marks)

QUESTION FOUR (15 MARKS)

a) Justify how African indigenous education relate to modern education in terms of content, methods and venues (15 marks)

QUESTION FIVE (15 MARKS)

a) Explain **five** reasons why Africans demanded for more literacy education between 1945- 1960 in Kenya (5 marks)

b) Discuss **five** ways through which competency base curriculum will address the challenge of unemployment in Kenya (10 marks)

END

KARATINA UNIVERSITY
PHY 214 CAT 1 (2022/2023)

a) Find V_{TH}, R_{TH} and the load current I_L flowing through and load voltage across the load resistor in the circuit in Fig 1 below using Thevenin's Theorem. (7 marks)

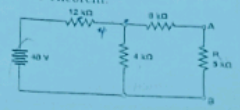


Figure 1

b) Using the circuit in figure 6 below:

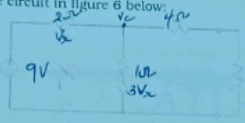
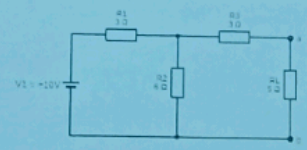


Figure 2

i. Determine the value of R_L that will draw the maximum power. (3 marks)
ii. Calculate the maximum power. (3 marks)

c) Calculate the current through the resistance RL = 5Ω by applying Norton's Theorem. (7 marks)



KARATINA UNIVERSITY
PHY 214 CAT 2 (2022/2023)

a) The figure 1 below shows an BJT made of germanium connected in common emitter mode.

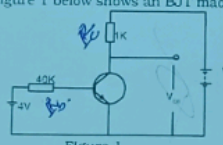


Figure 1

If the V_{cc}=12V and the the forward current gain β = 100. Calculate: I_b I_c , I_e V_{BE} , The value V_{CE} . (8 marks)

b) Show that the base current amplification factor, β is given by $\beta = \frac{\alpha}{1-\alpha}$ (2 marks)

c) Use superposition to find V_x in the circuit shown by figure 5 below. (6 marks)

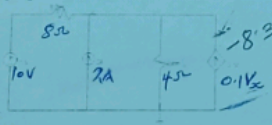


Figure 5

d) The figure 2 below shows a transistor which is operating in its active region

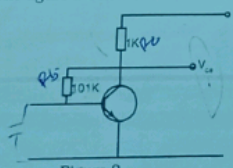


Figure 2

If Vcc=10.8V and β = 100. Find; I_B, I_C, and V_{CE} (4 marks)

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS
2022/2023ACADEMIC YEAR

SECOND YEAR **SECOND** SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE, EDUCATION SCIENCE AND SCIENCE WITH EDUCATION

COURSE CODE: PHY 214

COURSE TITLE: ANALOGUE ELECTRONICS

DATE: 18TH APRIL, 2023 TIME: 9.00 – 12.00NOON

INSTRUCTION TO CANDIDATES

SEE INSIDE

This paper has 4 printed pages.

INSTRUCTIONS:

- Answer all the questions
- Take: Mass of an electron $M_e=9.11x10^{-31}kg$, Electronic charge $e=1.6x10^{-19}C$, Acceleration due to gravity $g=9.8ms^{-2}$, Permittivity of free space $\epsilon_o=8.85x10^{-12}C^2m^{-2}N^{-1}$

Question one (25 Marks)

a) **Find V_{TH}, R_{TH} and the load current I_L flowing through load resistor in the circuit in Fig 1 below using Thevenin Theorem.** (6 marks)

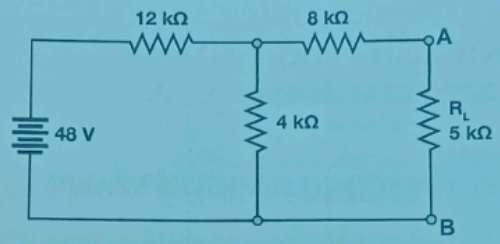


Figure 1

b) State the different between intrinsic and extrinsic semiconductors citing one example of each. (2 marks)

c) Using band semiconductors theory State, the differences between Conductor and semiconductor (2 marks)

d) Calculate the current through the resistance RL = 5Ω in Fig 2 by applying Norton's Theorem (5marks)

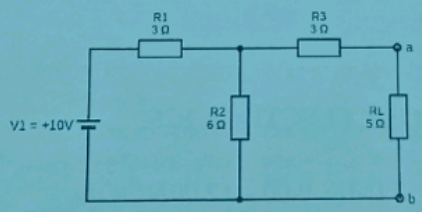


Figure 2

e) Find the current flowing through 20 Ω in Fig 3 using the superposition theorem.(5marks)

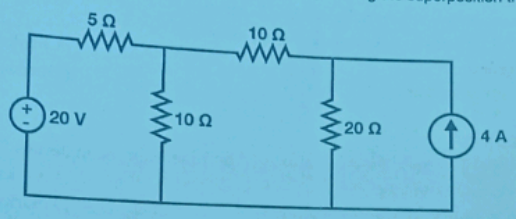


Figure 3

f) Using a diagram, Explain how an n-p-n Bipolar junction transistor circuit in common emitter mode can be used as an ac amplifier (5 Marks)

Question two (15 Marks)

a) Show that the relationship between β and α is $\beta = \frac{\alpha}{i-\alpha}$ (4 Marks)

b) The figure 4 below shows a BJT connected in common emitter mode. The following values are given;

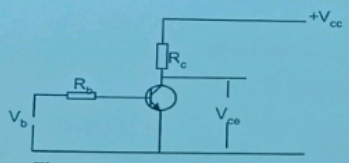


Figure 4

R_b=100K, R_c=4K, V_{cc}=10V, forward current gain β=100 and V_b= 2V. Determine:

i. The Base current I_b. (3marks)
ii. The collector Current I_c (3 marks)
iii. The emitter current Ie (2 marks)
iv. The Value of output Voltage Vce (3 marks)

Question three (15 marks)

a) Define the term doping as used in semiconductors devices. (1mark)
b) Explain any three characteristics of a common emitter amplifier (3marks)
c) With aid of a diagram explain the working of a bridge rectifier (5 marks)
d) With aid of well labelled diagrams explain the operations of
   i. inverting and (3 marks)
   ii. non-inverting op-amp (3 marks)

Question four (15 Marks)

a) Explain how the Light Emitting Diode is fabricated and mention the materials commonly used during fabrication. (4 marks)
b) Use superposition to find V_x in the circuit shown by figure 5 below. (5 marks)

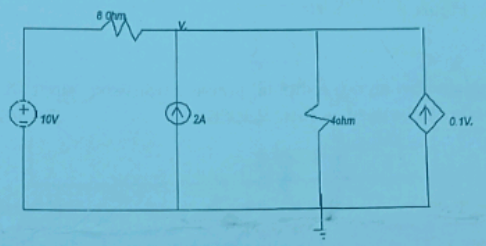


Figure 5

c) Using the circuit in figure 6 below:

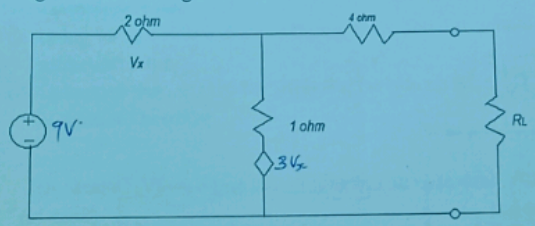


Figure 6

i. Determine the value of R_L that will draw the maximum power. (3 marks)
ii. Calculate the maximum power. (3 marks)

INSTRUCTIONS:

- Answer all the questions
- Take: Mass of an electron $M_e=9.11\times10^{-31}$kg, Electronic charge $e=1.6\times10^{-19}$C, Acceleration due to gravity $g=9.8ms^{-2}$.
- Permittivity of free space $\epsilon_o=8.85\times10^{-12}C^2m^{-2}N^{-1}$

Question one (25 Marks)

a) Explain how the Light Emitting Diode is fabricated and mention three materials commonly used during fabrication. (5 marks)

b) Show that the relationship between β and α is $\beta = \frac{\alpha}{i-\alpha}$ (3 Marks)

c) With aid of well labelled diagrams explain the operations of
   i. inverting and (2 marks)
   ii. non-inverting op-amp (2 marks)

d) Define the term doping as used in semiconductors devices. (1mark)

e) State the difference between intrinsic and extrinsic semiconductors citing two examples of each. (3 marks)

f) Using band semiconductors theory:
   i. State two similarities between Insulators and semiconductors (2 marks)
   ii. State two differences between Conductor and semiconductor (2 marks)

g) With aid of a well labelled diagram, explain how the bridge rectifier is used to convert ac into dc (5marks)

Question two (15 Marks)

a) State three differences between class A and Class B amplifiers (3 marks)

b) Use the superposition theorem to find the value of V in the circuit in figure 1 below. (4marks)

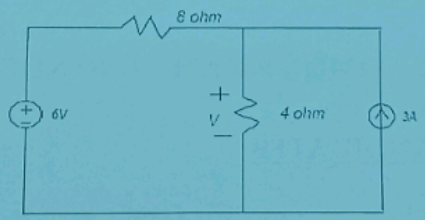


Figure 1

c) Find the Thevenin equivalent circuit at the terminals a & b shown in figure 2 below (4 marks)

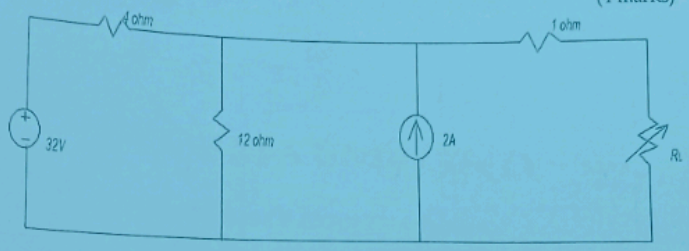


Figure 2

d) State any four characteristics of a common emitter amplifier (4marks)

Question three (15 marks)

a) Explain how an n-p-n Bipolar junction transistor circuit in common emitter mode can be used as an ac amplifier (4 Marks)

b) The figure 3 below shows a BJT connected in common emitter mode. The following values are given;

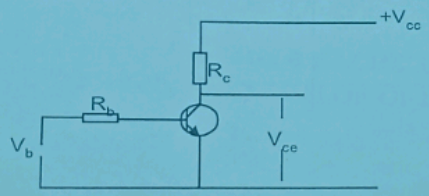


Figure 3

R_b=100K, R_c=4K, V_{cc}=10V, forward current gain β=100 and V_b= 2V. Further given that $\alpha=I_c/I_e$ determine:

i. The Base current I_b. (3marks)

ii. The collector Current I_c (3 marks)

iii. The emitter current Ie (2 marks)

iv. The Value of output Voltage Vce (3 marks)

Question four (15 Marks)

a) Find the Norton equivalent circuit with dep. Source for the circuit shown in the figure 4 below. (5 marks)

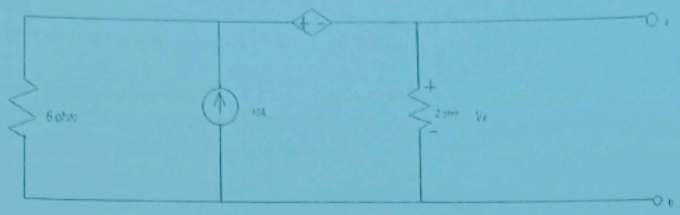


Figure 4

b) Use superposition to find V_x in the circuit shown by figure 5 below. (4 marks)

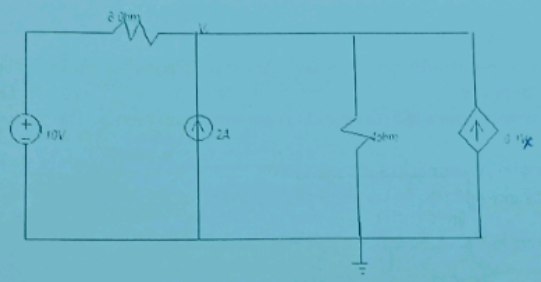


Figure 5

c) Using the circuit in figure 6 below:

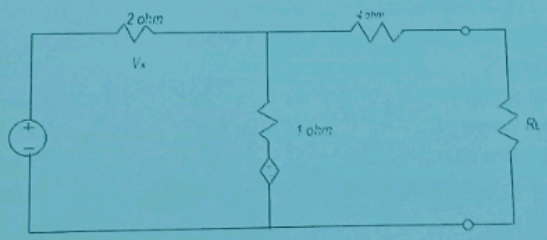
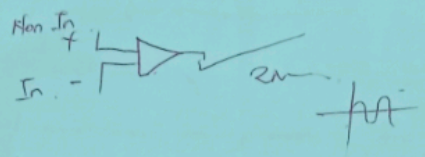


Figure 6

i. Determine the value of R_L that will draw the maximum power. (3 marks)

ii. Calculate the maximum power. (3 marks)



1. Answer all questions in section A and B
2. Write all your answers in the answer booklet provided
3. Time allowed: **THREE** Hours

E 11 3a1

SECTION A (25 MARKS)
QUESTION ONE

(a) With reference to electronics, write short notes giving examples on following terms:
(i) Biasing [2 marks]
(ii) Doping [2 marks]
(iii) Oscillator [2 marks]
(iv) Feedback [2 marks]
(b) Colpitts Oscillator circuit having two capacitors of 24nF and 240nF respectively are connected in parallel with an inductor of 10mH.
i) Determine the frequency of oscillations of the circuit [2 marks]
ii) Calculate the feedback fraction. [2 marks]
(c) Illustrates using diagram how common base transistor works [4 marks]
(d) Explain using diagram Wien Bridge Oscillator Circuit and it's working [6 marks]
(e) A crystal is doped with impurity atoms having valence electrons in the outer shell.
i) How will this affect the resistance of the semiconductor material and the current that flows through it? [2 marks]
ii) A n-type semiconductor material is formed by doping a pure semiconductor with impurity atoms. What is meant by the term n-type semiconductor material? [1 marks]

SECTION B (45 MARKS)
QUESTION TWO

(a) Differentiate oscillator from amplifier. [4 marks]
(b)With help of diagrams show Inverting and Non-Inverting Amplifier, [2 marks]
(c) Differentiate between unilateral and bilateral amplifiers [3 marks]
(d) Materials can be classified as conductors, semiconductors and insulators.
i) Give two examples of each material listed above. [3 marks]
ii) Why insulators is a bad conductor [2 marks]
(e) Derive a relationship between alpha and beta: [1 mark]

QUESTION THREE

(a) Explain conditions for a circuit to maintain oscillations [2 marks]
(b) Explain the advantages and disadvantages of RC phase shift oscillators? [6 marks]
(c) How do pentavalent impurities differ from trivalent impurities? Name two elements in each group [4 marks]
(d) With the help of a neat labelled diagram explain I-V Characteristics of a diode [3 marks]

QUESTION FOUR

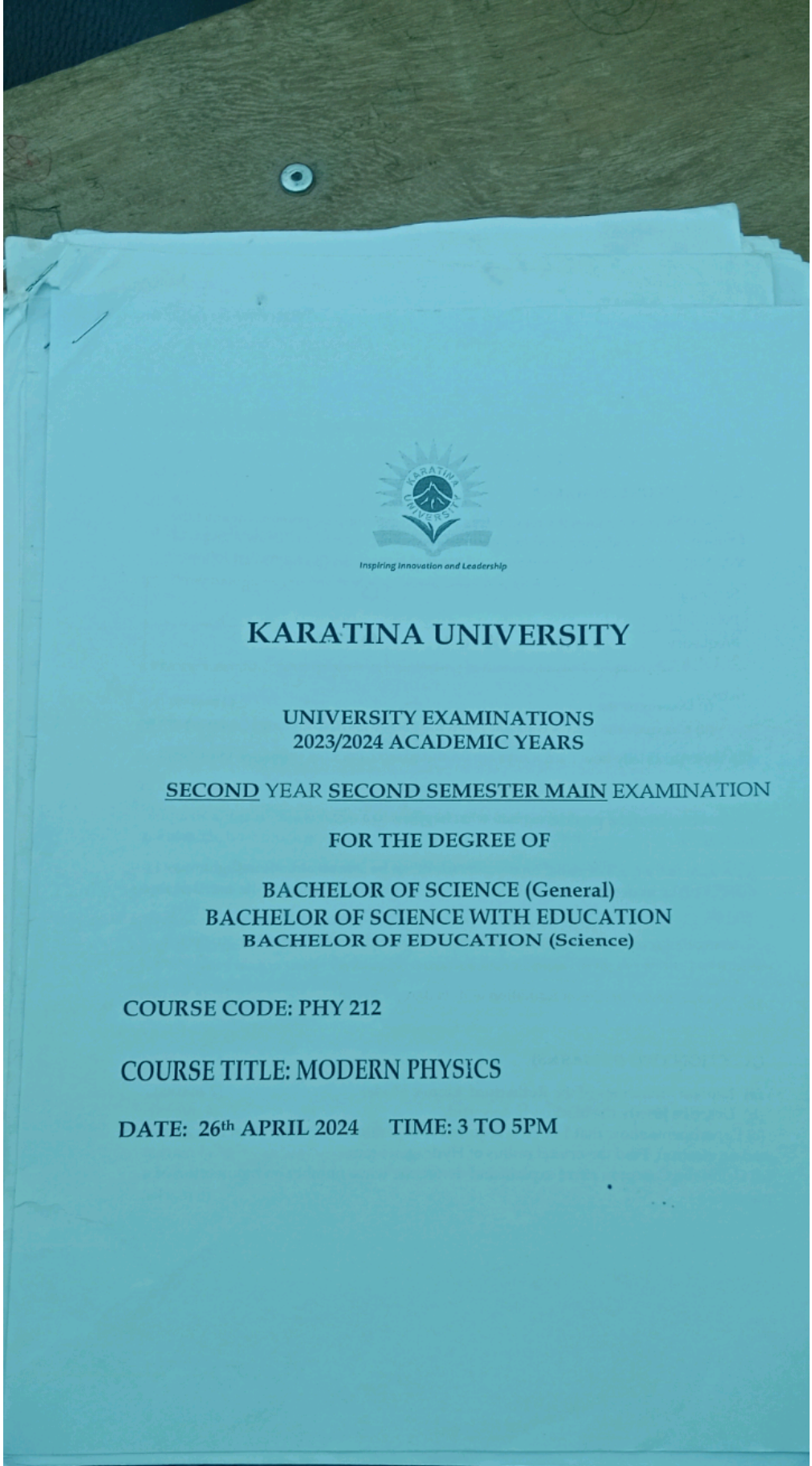
(a) With a neat diagram explain class AB amplifier, highlights its advantages and disadvantages [6 marks]

(b) In a common base connection, $I_E = 1mA$, $I_C = 0.95mA$. Calculate the value of I_B. [2 marks]

(c) Explain the procedure and steps in order to find the response in a particular branch using superposition theorem. [3 marks]

(d) Inverting input is a 'virtual ground' in op amp. What does it mean and what is its significance?. [2 marks]

(e) Distinguish between potential difference and depletion layer [2 marks]



Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS
2023/2024 ACADEMIC YEARS

SECOND YEAR SECOND SEMESTER MAIN EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE (General)
BACHELOR OF SCIENCE WITH EDUCATION
BACHELOR OF EDUCATION (Science)

COURSE CODE: PHY 212

COURSE TITLE: MODERN PHYSICS

DATE: 26th APRIL 2024 TIME: 3 TO 5PM

QUESTION ONE (25 MARKS)

(a)The table below shows the relationship between the stopping potential against the frequency of the radiation used in a photoelectric effect experiment. Draw a graph of stopping potential against frequency and use it to answer the questions that follow.

Stopping potential (v)	0.5	1.0	1.5	2.0
Frequency $\times 10^{14} Hz$	4.6	6.5	8.5	10.0

(i) Determine the threshold wavelength of the radiation used. (3 marks)

(ii) Calculate the planks constant given the known electronic charge. (4 marks)

(b) Describe briefly how hard x rays are produced from an x ray tube. (3 marks)

(c) Explain limitations of the Rutherford Atomic Model (2 marks)

(d) Describe an alpha particle; explain what happens to a nucleus after it emits an alpha particle (3 marks)

(e) A man has a mass of 100kg on the ground. When he is in an aircraft in flight, his mass is 101kg as determined by an observer on the ground. Calculate the speed of the aircraft. (3 marks)

(f) Using diagram explain photoelectric effect, explain characteristics of photoelectric effect (3 marks)

(g) Explain the Properties of Radiation and its danger in life (4 marks)

QUESTION TWO (15 MARKS)

(a) Explain limitations of the Rutherford Atomic Model (2 marks)

(b) Describe briefly the blackbody spectrum. (4 marks)

(c) Experiments show that 13.6eV is required to separate a hydrogen atom into a proton and an electron. Find the orbital radius of Hydrogen atom. (3 marks)

(d) Using the Compton effect explain and derive the wave number on momentum of a photon (6 marks)

QUESTION THREE (15 MARKS)

(a) Electrons are ejected from a surface with speeds ranging up to 770km/s when radiation of wavelength 392nm illuminates the surface.

(i) What is the work function of the material [2 marks]

(ii) What is the threshold frequency for the surface [2 marks]

(iii) What is the stopping potential for the surface [2 marks]

(b) Show that the wavelength of a scattered photon is given by

$$\lambda' = \lambda + \frac{h}{m_e c}(1 - \cos\theta)$$

(6 marks)

(c) Using neat diagram explain x-ray scattering in the relation to Compton effect

(3 marks)

QUESTION FOUR (15 MARKS)

(a) Derive the equation for length contraction (6 marks)

(b) In a Compton collision with an electron, a photon of violet light (λ= 400nm) is backscattered through an angle of 180^{0}.

i) Calculate how much energy (eV) is transferred to the electron (2 marks)

ii) Determine energy that would an electron acquire in a photoelectric process with the same photon (2 marks)

(c)Define nuclear binding energy and give its expression. (2 marks)

(d) Estimate the temperature of a surface of the sun given that the sun emits light whose peak intensity occurs in the visible spectrum at around 500nm. State the assumption made when solving the problem. (3 marks)

QUESTION ONE (25 MARKS)

(a)The table below shows the relationship between the stopping potential against the frequency of the radiation used in a photoelectric effect experiment. Draw a graph of stopping potential against frequency and use it to answer the questions that follow.

Stopping potential (v)	-0.5	1.0	1.5	2.0
Frequency $\times 10^{14} Hz$	0	7.2	8.6	10.0

(i) Determine the threshold wavelength of the radiation used. (3 marks)

(ii) Calculate the planks constant given the known electronic charge. (4 marks)

(b) Describe briefly how hard x rays are produced from an x ray tube. (3 marks)

(c) State two postulates of special relativity. (2 marks)

(d) Two observers, P on earth and Q in a spacecraft whose speed is $2x10^8 m/s$, both set their watches to the same time when the spacecraft is abreast of the earth. Determine the time that elapses by P's reckoning before the watches differ by 1s. (3 marks)

(e) A man has a mass of 100kg on the ground. When he is in an aircraft in flight, his mass is 101kg as determined by an observer on the ground. Calculate the speed of the aircraft. (3 marks)

(f) State three uses of photoelectric effect. (3 marks)

(g) An electron is accelerated through a potential difference of 40kV. Calculate its de Broglie wavelength. (3 marks)

(h) State Paulli's Exclusion principle (1mark)

QUESTION TWO (15 MARKS)

(a) state two examples of massless particles. (2 marks)

Show that for massless particles ($m_o=0$) the relation between their energy, momentum and velocity is given by $E = Pc$. (5 marks)

(b) Describe briefly the blackbody spectrum. (5 marks)

(c) Experiments show that 13.6eV is required to separate a hydrogen atom into a proton and an electron. Find the orbital radius of Hydrogen atom. (3 marks)

QUESTION THREE (15 MARKS)

(a) The work function of Caesium surface is 1.94eV. A light source of $3.8 \times 10^{-7} m$ wavelength is directed to the surface of Caesium surface. Determine

(i) Threshold frequency. (3 marks)

(ii) Maximum kinetic energy of the photoelectron. (3 marks)

(b) Determine decay constant of a radio-active element whose half life is 3.82 days. (3 marks)

(c)) Show that the wavelength of a scattered photon is given by

$$\lambda' = \lambda + \frac{h}{m_e c}(1 - \cos\theta)$$

(6 marks)

QUESTION FOUR (15 MARKS)

(a) Derive the equation for length contraction (6 marks)

(b)A metre stick moves with velocity of 0.6C relative to the observer along the direction of its length. Determine the time it will take for the metre stick to pass the observer. (4 marks)

(c)Define nuclear binding energy and give its expression. (2 marks)

(d) Estimate the temperature of a surface of the sun given that the sun emits light whose peak intensity occurs in the visible spectrum at around 500nm. State the assumption made when solving the problem. (3 marks)

ANSWER ALL QUESTIONS

SECTION A (25 MARKS)

(a) Define the term natural radioactivity (2 marks)

(b) Briefly explain the phenomenon referred as neutrino oscillation (2 marks)

(c) State the Galilean transformation coordinates (2 marks)

(d) Name any three bosons in the standard model of particle physics (3 marks)

(e) Briefly describe any two consequences of special relativity (4 marks)

(f) State Rayleigh-Jeans law (1 mark)

(g) Show that for large wavelengths, Rayleigh Jeans law reduces to planks law (3marks)

(h) State the special theory of relativity. (1 mark)

(i) Distinguish between nuclear fision and nuclear fusion. (2 marks)

(j) A stick 1m high appears 80cm tall to an observer. Determine the relative speed and how long it takes to pass the observer. (3 marks)

(k) State Einstein's second postulate of special relativity (1mark)

(l) State Pauli's exclusive principle (1 marks)

QUESTION TWO (15 MARKS)

(a) From first principles derive Einstein's mass energy equation, $E = mc^2$ (5 marks)

(b) Distinguish between bremsstrahlung and characteristic x-rays (2 marks)

(c) Prove that for non-relativistic motion, Lorentz transformation coordinates reduce to the classical transformation coordinates (4 marks)

(d) Determine the approximate number of photons emitted per second by a 5.00-MW green laser pointer (assume the wavelength of green light is 532 nm)(4 marks)

QUESTION THREE (15 MARKS)

(a) Experiments show that 13.6eV is required to separate a hydrogen atom into a proton and an electron.

(i) Find;

(I) Orbital radius (3 marks)

(II) Velocity of the electron in the hydrogen atom. (3 marks)

(ii) Hence show that the orbit of the electron in a hydrogen atom corresponds to one complete electron wave joined onto itself. (4 marks)

(b) With an aid of a well labeled diagram, describe briefly Rutherford scattering experiment. (5 marks)

QUESTION FOUR (15 MARKS)

(a) Derive the exponential radioactive decay law (5 marks)

(b) Briefly describe how carbon dating is used to measure the age of fossils as used in archeological studies. (4 marks)

(c) Show that the wavelength of a scattered photon is given by

$$\lambda' = \lambda + \frac{h}{m_e c}(1 - \cos\theta)$$ (6 marks)

Volume of acid = 28.70 cm³

INSTRUCTIONS

Attempt question one and any other three questions

Useful constants

Electron charge, $e=1.6\times10^{-19}$ C
Speed of light $c=2.99\times10^{8}$ ms^{-1}
Plank's constant, $h=6.62\times10^{-34}$ J.s
Electron mass, $m_e=9.11\times10^{-31}$ kg
Rest mass energy of an electron =0.511 MeV
Q value for ^{235}U=208 MeV
Wavelength of green light = 532 nm

Question one

(a) Explain the term covariant as used in relativistic mechanics (1 mark)
(b) State Einstein's firstpostulate of special relativity (1mark)
(c) The period of a pendulum is measured to be 5.0 s in the rest frame of the pendulum Determine the period of the pendulum when measured by an observer moving at a speed of 0.88c with respect to the pendulum. (4 marks)
(d) Explain why the average energy of the photoelectrons generated by a faint violet light is higher than that of those by a bright green light. (2marks)
(e) State Heisenberg's uncertainty principle (2marks)
(f) Describe the nuclear shell theory (3 marks)
(g) Calculate the de Broglie wavelength of an electron accelerated through a 100V potential difference. (4marks)
(h) Explain the term natural radioactivity (2 marks)
(i) Prove that for non-relativistic motion, Lorentz transformation coordinates reduce to the classical transformation coordinates (4 marks)
(j) State Pauli's exclusive principle (2 marks)

Question two

(a) Determine the approximate number of photons emitted per second by a 5.00-mW green laser pointer. (4 marks)
(b) By considering an observer holding a laser source inside a moving vehicle with a reflecting surface at the mounted on the ceiling, determine the expression for proper time taken for the light from the source to be reflected back (4 marks)

(c) Calculate the total energy released if 3.5kg of 235U undergoes fission (3 marks)
(d) Derive the exponential decay law (4 marks)

Question three

(a) Describe the Compton scattering process (2 marks)
(b) An X-ray with a frequency of 3.3530×10^{19} Hz strikes a metal foil and the scattered photon is detected at an angle of 32.3 degrees relative to the direction of the original X-ray.
(i) Determine the energy of the incoming photon and that of the scattered photon in electron volts.(6 marks)
(ii) Determine the kinetic energy and momentum of the electron after collision (4 marks)
(c) Show that for large wavelengths, plank's distribution reduces to Rayleigh-Jeans distribution (3 marks

Question four

(a) Suppose you have an electron and an alpha particle moving through a piece of beam pipe inside a particle accelerator. The particles are traveling in the same direction. The speed of the electron is 0.830*c* and the speed of the alpha particle is 0.750*c*, both measured by a stationary observer in the lab.Determine the speed of the alpha particle as observed from the electron. (4 marks)
(b) Name any two bosons in the standard model of particle physics (2 marks)
(c) Show that the relativistic kinetic energy of a particle is given by

$$K = \gamma mc^2 - mc^2$$

where m is the proper mass of the particle (6 marks)
(d) Write the electron configurations for hydrogen, Beryllium and Boron (3 marks)

Useful constants

Electron charge, $e=1.6\times10^{-19}$ C

Speed of light $c=2.99\times10^{8}$ ms^{-1}

Plank's constant, $h=6.62\times10^{-34}$ J.s

Electron mass, $m_e=9.11\times10^{-31}$ kg

Rest mass energy of an electron =0.511 MeV

Q value for ^{235}U=208 MeV

Question one

(a) Explain the term covariant as used in relativistic mechanics (2 marks)

(b) State Einstein's second postulate of special relativity (1marks)

(c) By considering an observer holding a laser source inside a moving vehicle with a reflecting surface at the mounted on the ceiling, determine the expression for proper time taken for the light from the source to be reflected back (4 marks)

(d) Explain why the average energy of the photoelectrons generated by a bright blue light is lower than that of those generated by a faint violet light. (2marks)

(e) State the main application of absorption spectroscopy in research (1marks)

(f) By using the idea of standing waves, verify Bohr's assumption of quantized angular momentum (4marks)

(g) Briefly describe the nuclear shell theory (3 marks)

(h) Calculate the de Broglie wavelength of an electron accelerated through a 125V potential difference. (3 marks)

(i) Define the the term artificial radioactivity (2 marks)

(j) Briefly explain the phenomenon referred as neutrino oscillation (2 marks)

Question two

(a) Determine the approximate number of photons emitted per second by a 5.00-mW green laser pointer (assume the wavelength of green light is 532 nm) (5 marks)

(b) State the Galilean transformation coordinates (2 marks)

(c) Derive the exponential radioactive decay law (5 marks)

$$x' = (x - vt)$$
$$y' = y$$
$$z' = z$$
$$t' = t$$

$$x' = \gamma(x - vt)$$
$$y' = y$$
$$z' = z$$
$$t' = \gamma\left(t - \frac{vx}{c^2}\right)$$

$$x = \gamma(x' + vt')$$
$$y = y'$$
$$z = z'$$
$$t = \gamma\left(t' + \frac{v'_x}{c^2}\right)$$

Question three

(a) Distinguish between bremsstrahlung and characteristic x-rays (2 marks)

(b) Show that the wavelength of a scattered photon is given by

$\lambda' = \lambda + \frac{h}{m_e c}(1-\cos\theta)$ (6 marks)

The symbols have their usual meaning

(c) Prove that for non-relativistic motion, Lorentz transformation coordinates reduce to the classical transformation coordinates (4 marks)

Question four

(a) From first principles derive Einstein's mass energy equation, $E = mc^2$ (6marks)

(b) Suppose you have an electron and an alpha particle moving through a piece of beam pipe inside a particle accelerator. The particles are traveling in the same direction. The speed of the electron is 0.830c and the speed of the alpha particle is 0.750c, both measured by a stationary observer in the lab and c is the speed of light. Determine the speed of the alpha particle as observed from the electron.(3marks)

(c) Name any three bosons in the standard model of particle physics (3 marks)

Question five

(a) Find the kinetic energy of an electron moving at a speed of 0.75c where is the speed of light (4 marks)

(b) State Pauli's exclusive principle (1 marks)

(c) Briefly discuss the positron emission tomography technique (3 marks)

(d) Write the electron configurations for Hydrogen, Helium , beryllium and Boron (4 marks)

Question six

(a) State Rayleigh-Jeans law (1 mark)

(b) Show that for large wavelengths, Rayleigh Jeans law reduces to planks law (3marks)

(c) Calculate the total energy released if 3.5kg of ^{235}U undergoes fission (4 marks)

(d) Briefly describe any two consequences of special relativity (4 marks)

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS

2023/2024 ACADEMIC YEAR

SECOND YEAR SECOND SEMESTER REGULAR EXAMINATIONS

FOR THE DEGREE OF:

BACHELOR OF SCIENCE (P102,P103,P106,P107) AND BACHELOR OF EDUCATION (E100,E101,E103,E111,E112)

COURSE CODE: MAT 221

COURSE TITLE: ORDINARY DIFFERENTIAL EQUATIONS I

DATE: 25th APRIL, 2024 TIME: 3:00 PM - 5:00 PM

Instructions: See Inside

Answer **all** questions in section A and any other **two** from section B.

SECTION A: COMPULSORY (30 MARKS)

Answer **all** questions from this section

QUESTION ONE (30 Marks)

(a) Use the method of solution by integration to solve the given ordinary differential equations

$$\frac{dy}{dx} = \frac{xy^2 + x}{yx^2 + y}$$

[4 Marks]

(b) Solve the given differential equation using the method of substitution

$$2xy\frac{dy}{dx} = y^2 - x^2$$

[5Marks]

(c) A circuit in series has a constant electromotive force of 40V, a resistance of 10Ω, an inductance of 0.2 henry and an initial current of 0. Solve the basic differential equation to find the current i on the circuit at any time $t > 0$ [5 Marks]

(d) Solve the given differential equation using reduction by order method

$$y'' + 4y = 0, \qquad y_1 = \cos 2x$$

[4 Marks]

(e) Solve the given Bernoulli differential equation

$$x\frac{dy}{dx} + y = x^3y^6$$

[5 Marks]

(f) Find the solution to the given Non-Homogeneous Linear Equations of Second Order

$$y'' + 3y' + 2y = 4x^2$$

[7 Marks]

SECTION B: ELECTIVE (40 MARKS)

Answer **any Two** questions from this section

QUESTION TWO (20 Marks)

(a) Show that

$$3x(xy - 2)dx + (x^3 + 2y)dy = 0$$

is an exact differential,hence solve the equation. [5 Marks]

(b) Suppose that a pie is removed from $350°F$ oven and placed in a room with a temperature of $75°F$. In 15 minutes, the pie has a temperature of $150°F$ Determine the time required to cool the pie to a temperature of $80°F$ when one can actually enjoy the eating? [5 Marks]

(c) Using the method of variation of parameters evaluate

$$y'' - 4y' + 3y = e^{-x}$$

[10 Marks]

QUESTION THREE (20 Marks)

(a) Use substitution method to solve the special case of homogeneous differential equation given as

$$\frac{dy}{dx} = \frac{x + y + 4}{x - y - 6}$$

[10 Marks]

(b) Find solution to the given Non-Homogeneous Linear Equations of Second Order with Constant Coefficients expressed as

$$y'' - 5y' + 6y = -3\sin 2x$$

[10 Marks]

QUESTION FOUR (20 Marks)

(a) Solve the initial value problem

$$x\frac{dy}{dx} = y + \sqrt{x^2 - y^2} \qquad y(x_0) = 0 \qquad x_0 > 0$$

[6 Marks]

(b) Using the method of variation of parameters compute the solution to

$$y'' + y = \sec x$$

[8 Marks]

(c) Solve the given Cauchy-Euler Differential Equation

$$4x^2y'' + 8xy' + y = 0$$

[6 marks]

QUESTION FIVE (20 Marks)

(a) Find the general solution to

$$\frac{1}{x}\frac{dy}{dx} - \frac{2y}{x^2} = x\cos x \quad x > 0$$

[4 Marks]

(b) Suppose a population has 39 members at time $t = 8$ and 60 members at time $t = 12$. Find the original population [5 Marks]

(c) Find the particular integral of

$$\frac{d^2y}{dx^2} + y = \csc x$$

[6 Marks]

(d) Solve the differential equation using the method of variable separable

$$\frac{dy}{dx} = (9x + y + 1)^2$$

[5 Marks]

Answer **all** questions in section A and any other **three** from section B.

SECTION A

Answer **all** questions from this section

QUESTION ONE (15 marks)

(a) Solve the given Bernoulli differential equation

$$x\frac{dy}{dx} + y = x^3y^6$$

[5 Marks]

(b) Solve the given differential equation using the method of substitution

$$2xy\frac{dy}{dx} = y^2 - x^2$$

[4 Marks]

(c) Solve the given initial value problem

$$(e^{2y} - y)\cos x\frac{dy}{dx} = e^y \sin 2x, \quad y(0) = 0$$

[3 Marks]

(d) A colony of bacteria increases according to the law of exponential growth. If the number of bacteria doubles in 3 hours. How long will it take for the size of the colony to triple.

[3 Marks]

QUESTION TWO (16 marks)

(a) Find the solution to the given Non-Homogeneous Linear Equations of Second Order

$$y'' + 3y' + 2y = 4x^2$$

[8 Marks]

(b) Show that

$$y = e^{\alpha x}[A\cos\beta x + B\sin\beta x]$$

whenever

$$m_1 = \alpha + i\beta \quad m_2 = \alpha - i\beta$$

[4 Marks]

(c) Verify that

$$y'' - 2y' + y = 0$$

is the solution to $y = xe^x$ on the interval $(-\infty, \infty)$ [3 Marks]

SECTION B

Answer **any Three** questions from this section

QUESTION THREE (13 marks)

(a) Solve the exact differential equation given

$$2xy\,dx + (x^2 - 1)\,dy = 0$$

[5 Marks]

(b) A circuit in series has a constant electromotive force of 40V, a resistance of 10Ω, an inductance of 0.2 henry and an initial current of 0. Solve the basic differential equation to find the current i on the circuit at any time $t > 0$ [5 Marks]

(c) Solve

$$y'' - y' - 6y = 0$$

[3 Marks]

QUESTION FOUR (13 marks)

Using the method of variation of parameters evaluate

$$y'' - 4y' + 3y = e^{-x}$$

[13 Marks]

QUESTION FIVE (13 marks)

(a) Find solution to the given Non-Homogeneous Linear Equations of Second Order with Constant Coefficients expressed as

$$y'' - 5y' - 6y = -3\sin 2x$$

[7 Marks]

(b) Solve the given differential equation using reduction by order method

$$y'' + 4y = 0, \qquad y_1 = \cos 2x$$

$y_2 = y_1 \int \frac{e^{\int p(x)dx}}{y_1^2}$

[5 Marks]

QUESTION SIX (13 marks)

(a) Use substitution method to solve the special case of homogeneous differential equation given as

$$\frac{dy}{dx} = \frac{x+y+4}{x-y-6}$$

[8 Marks]

(b) Solve the differential equation using the method of variable separable

$$\frac{dy}{dx} = (9x+y+1)^2$$

[5 Marks]

QUESTION SEVEN (13 marks)

(a) Suppose that a pie is removed from 350^oF oven and placed in a room with a temperature of 75^oF. In 15 minutes, the pie has a temperature of 150^oF Determine the time required to cool the pie to a temperature of 80^oF when one can actually enjoy the eating? [4 marks]

(b) Find the general solution to

$$\frac{1}{x}\frac{dy}{dx} - \frac{2y}{x^2} = x\cos x \quad x > 0$$

[3 Marks]

(c) Find the particular integral of

$$\frac{d^2y}{dx^2} + y = \csc x$$

[6 Marks]

INSTRUCTIONS: Answer ALL questions in SECTION A and any other THREE in SECTION B

SECTION A:(31 marks)

QUESTION ONE (16 MARKS)

a. Citing an example, define an ordinary differential equation. (2 marks)

b. Powdered milk is being transformed to liquid milk at a rate that is directly proportional to the unconverted milk. If originally there was 30kg of powdered milk. Form the ODE describing this information. (4 marks)

c. Find the particular integral $\frac{d^2y}{dx^2}+4y=\sin 2x$ (3 marks)

d. Show that the differential equation $(2xy+\cos y)dx+(x^2-x\sin y)dy=0$ is exact. Hence, find its general solution. (4 marks)

e. Show that $x^2+y^3=cx$ is the solution of the differential equation,

$$x^2+3xy^2\frac{dy}{dx}=y^3$$ (3 marks)

QUESTION TWO (15 MARKS)

a. Solve the differential equation $(x-2y)dx+xdy=0$ (4 marks)

b. Find the general solution of $\frac{d^2y}{dx^2}+6\frac{dy}{dx}+13y=0$ (3 marks)

c. Show that $y=e^{2x}$ and $y=e^{-3x}$ are both the solutions to the differential equation defined by $\frac{d^2y}{dx^2}+\frac{dy}{dx}-6y=0$ (4 marks)

d. Obtain the differential equation associated with the primitive $y=Ax^2+Bx$, where A and B are essential arbitrary constants. (4 marks)

SECTION B (39 marks)

QUESTION THREE (13 MARKS)

a. Solve the following differential equation $xdy+(2xy-x)dx=0$ (5 marks)

$x\,dy+(2xy-x)\,dx=0$

$\frac{x\,dy}{2}+\frac{x(2y-1)\,dx}{2y-1}=0$

$\frac{1}{x}\left(\frac{x\,dy}{2y-1}+\frac{x\,dx}{x}\right)=0$

$\frac{dy}{2y-1}+dx=0$

$[2y-1]^{-1}dy=dx$

$\int\left(\frac{1}{2y-1}\right)dy=\int -dx$

b. Find the complete solution of $\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=\sin 2x$. (6 marks)

c. Given the equation $(y-x^2-1)dx+xdy=x$;

show that the equation can be put in the form $\frac{dy}{dx}+yP(x)=Q(x)$. (2marks)

QUESTION FOUR (13 MARKS)

Prove that the transformation $z=y^{1-n}$ reduces the equation $\frac{dy}{dx}+P(x)y=Q(x)y^n$ to a linear differential equation in z and x. Hence, solve the differential equation defined by $\frac{dy}{dx}+y=xy^4$. (13 marks)

QUESTION FIVE (13 MARKS)

a. Solve the initial value problem $\frac{d^2y}{dx^2}+6\frac{dy}{dx}+9y=0$ given that y(0)= - 4 and $y'(0)=14$ (3 marks)

b. Use the method of variation of parameters to find the complete solution of the following differential equations $\frac{d^2y}{dx^2}+y=\sec x$ (10 marks)

QUESTION SIX (13 MARKS)

a. Use the method of undetermined coefficients to find the complete solutions of the following differential equations $\frac{d^2y}{dx^2}-2\frac{dy}{dx}-3y=xe^{-x}$ (9 marks)

b. A curve is defined by the condition that at each point the gradient is equal to twice the sum of the coordinates. Find the equation of the curve. (4 marks)

QUESTION SEVEN (13 MARKS)

a. Differentiate between Order of a differential equation and Degree of differential equation, hence state order and degree of the differential equation given $\frac{dy}{dx}+y^2=\cos x$. (3 marks)

b. Find the general solution of $\frac{d^2y}{dx^2}-2\frac{dy}{dx}-3y=e^{-2x}\cos 3x$ (10 marks

INSTRUCTIONS: Answer **ALL** questions in **SECTION A** and any other **THREE** in **SECTION B**

SECTION A:

QUESTION ONE (16 MARKS)

a. Define an ordinary differential equation and give an example. (2 marks)

b. Differentiate between Order of a differential equation and Degree of differential equation, hence state order and degree of the differential equation given $\frac{dy}{dx} + y^2 = \cos x$ (4 marks)

c. Find the particular integral $\frac{d^2y}{dx^2} + 4y = \sin 2x$ (5 marks)

d. Show that the differential equation $(2xy + \cos y)dx + (x^2 - x\sin y)dy = 0$ is exact. Hence, find its general solution. (5 marks)

QUESTION TWO (15 MARKS)

a. Solve the differential equation $(x - 2y)dx + xdy = 0$ (4 marks)

b. Find the general solution of $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 13y = 0$ (3 marks)

c. Show that $y = e^{2x}$ and $y = e^{-3x}$ are both the solutions to the differential equation defined by $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$ (4 marks)

d. Obtain the differential equation associated with the primitive $y = Ax^2 + Bx$, where A and B are essential arbitrary constants. (4 marks)

SECTION B

QUESTION THREE (13 MARKS)

a. Solve the following differential equation $xdy + (2xy - x)dx = 0$ (7 marks)

b. Find the complete solution of $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \sin 2x$. (6 marks)

QUESTION FOUR (13 MARKS)

a. Prove that the transformation $z = y^{1-n}$ reduces the equation

$\frac{dy}{dx} + P(x)y = Q(x)y^n$ to a linear differential equation in z and x. Hence, solve

the differential equation defined by $\frac{dy}{dx} + y = xy^4$ (13 marks)

QUESTION FIVE (13 MARKS)

a. A curve is defined by the condition that at each point the gradient is equal to twice the sum of the coordinates. Find the equation of the curve. (3 marks)

b. Use the method of variation of parameters to find the complete solution of

the following differential equations $\frac{d^2y}{dx^2} + y = \sec x$ (10 marks)

QUESTION SIX (13 MARKS)

a. Use the method of undetermined coefficients to find the complete solutions of

the following differential equations $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = xe^{-x}$ (9 marks)

b. Solve the initial value problem $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0$ given that y(0)= - 4 and

$y'(0) = 14$. (4 marks)

QUESTION SEVEN (13 MARKS)

a. Powdered milk is being transformed to liquid milk at a rate that is directly proportional to the unconverted milk. If originally there was 30kg of powdered milk. Form the ODE describing this information. (3 marks)

b. Find the general solution of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = e^{-2x}\cos 3x$ (10 marks)

$10 = X_0 e^{-kt} \Rightarrow 10 = X_0 e^{-k}$

$5 = X_0 e^{-4k}$ $\quad 5 = X_0 e^{-3k}$

$\frac{10}{5} = \frac{X_0 e^{-k}}{X_0 e^{-3k}}$

$2 = \frac{e^{-k}}{e^{-3k}}$

$2e^{-3k} = e^{-k}$

$2 = \frac{e^{-k}}{e^{-3k}}$ $\quad \int \frac{1}{y}\, \ln|y|$

$2 = e^{-k} \cdot e^{3k}$

$2 = e^{-2k}$ $\quad 0.6931 = -2k$

$k =$

SECTION A

Answer Both Questions in this Section (31 Marks)

QUESTION ONE (15 marks)

a) Show that the substitution $z = ax + by$ transforms the following equation into one of separable variables and hence solve the equation.

$$\frac{dy}{dx} = \frac{1+2x+2y}{1-2x-2y}$$ (4 marks)

b) Determine if the following differential equation is exact or not.

$$(3x-2y-4)dx + (5y-2x+1)dy = 0$$

Hence solve the equation. (5 marks)

c) The rate at which a substance is transformed into another in chemical reaction is proportional to the amount untransformed. If x is the amount untransformed at time t, and $x = 10$ when $t = 1$ and $x = 5$ when $t = 3$, find the value of x at the start. (6 marks)

QUESTION TWO (16 marks):

a) (i) Define the term , ordinary differential equation (1 mark)

(ii) If $D = \frac{d}{dx}$ is the differential operator, write the equation $xy'' + 5y' - 3xy = \cos x$ in D operator form. (1 mark)

b) Show that $x^2 + y^2 = cx$ is the solution of the differential equation,

$$x^2 + 3xy^2\frac{dy}{dx} = y^3$$ (3 marks)

c) (i) Solve $(\sqrt{x^2+1})\frac{dy}{dx} = \frac{x}{y}$ by separation of variables. (3 marks)

(ii) Show that the equation $x^2\frac{dy}{dx} = 3x^2 + xy$ is homogeneous and hence solve it. (5 marks)

d) The distance S covered in time t by body is given by the equation $\frac{d^2S}{dt^2} = 24t^4 - 12t^2 + 6$

If at time $t = 0$ and $t = 1$ the distances covered are $S = 2$ and $S = 3$ respectively, determine $S = f(t)$ (3 marks)

SECTION B

Answer ANY Three Questions in this Section (39 Marks)

QUESTION THREE (13 Marks)

a) Find an integrating factor for the equation ; $(\frac{1}{x}\sin x + 3y^2)dx + 2xydy = 0$.

Hence use the integrating factor to solve the equation. (5 marks)

b) Given the equation $y'' + 2y' + 4y = 8x^2 + 12e^{-x}$,

(i) Determine the homogeneous solution (3marks)

(ii) Use the method of undetermined coefficients to find the particular solution (5 marks)

QUESTION FOUR (13 Marks).

a) If D^{-1} is the inverse differential operator, determine $(2+D)^{-1}(x^2+2x)$ (3 marks)

b) The differential equation $x\frac{dy}{dx} + 2y = y^2x^3$ is nonlinear.

(i) Explain why the equation is nonlinear (1 mark)

(ii) Use a suitable substitution to make it linear and then solve. (5 marks)

c) The acceleration, a and the velocity v, of a body sinking slowly in a liquid is given approximately by the equation $a = g - kv$ where g and k are constants.

Find the distance as a function of time, given that the body starts from rest. (4 marks)

QUESTION FIVE (13 Marks)

$y_1 = e^{m_1 x}$,
$y_2 = e^{m_2 x}$,

a) Given the equation $(y - x^2 - 1)dx + xdy = x$;

(i) show that the equation can be put in the form $\frac{dy}{dx} + yP(x) = Q(x)$ and (2 marks)

(ii) hence solve the equation using an integrating factor (4 marks)

b) Solve the differential equation, $(x-1)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$. (7 marks)

$m^2 - m$

$\frac{dx}{d} \times \frac{d}{dx}\left[(x-1)\frac{dy}{dx} - xy\right] = 0 \times \frac{dx}{d}$

$\frac{(x-1)}{x}\frac{dy}{dx} - \frac{xy}{x} = \frac{0}{x}$

$\frac{(x-1)}{x}\frac{dy}{dx} - y = 0$

$\int\left(\frac{x-1}{x}\right)\frac{dy}{dx} = y \times dx$

$\left(\frac{x-1}{x}\right)dy = y\,dx$

$\int\frac{1}{y}dy = \int\frac{x}{x-1}dx$

$\ln(y) = \tan^{-1}|x-1| +$

QUESTION SIX (13 Marks)

The differential equation, $a_2\frac{d^2y}{dx^2}+a_1\frac{dy}{dx}+a_0y=0$, where a_0, a_1 and a_2 are constants has a general solution given by $ye^{-m_1x}=c\int e^{(m_1-m_2)x}dx$ where m_1 and m_2 are the roots of the characteristic equation of the differential equation and c is an arbitrary constant.

(a) Write down the various the roots the solution can take (1 mark)

(b) Derive the solution for each case (8 marks)

(c) Use your results to solve the equation $9y''+6y'+10y=0$ (4 marks)

QUESTION SEVEN (13 Marks).

a) Use an operator method to determine a particular solution of the differential equation;

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}-3y=xe^x$$ (7 marks)

b) Differential equation $(D^2+2D-8)y=0$ has the solution $y=C_1e^{-4x}+C_2e^{2x}$.

Determine the solution of the following differential equation using the method of the variation of parameters;

$(D^2+2D-8)y=8x+4$ (6 marks)

$\frac{5}{10}=\frac{X_0e^{-3k}}{X^0e^{-k}}$

$-0.6931=e^{-2k}$

$\frac{(x-2y)dx}{dx}=\frac{xdy}{dx}$

$x-2y=\frac{xdy}{dx}-x$

$-2y=$

$(x-2y)\frac{dx}{dy}=-xdy$

Yac

$(x-2y)dx=-xdy$

$\frac{xdy}{x}=-\frac{(x-2y)}{x}dx$

$\int dy=\int(-1-\frac{2y}{x})dx$

$y=\int -1\,dx-\int\frac{2y}{x}d$

$y=-x-2y\int\frac{1}{x}dx$

$y=-x-2y[\ln|x|]$

SECTION A

Answer Both Questions in this Section (31 Marks)

QUESTION ONE (16 marks)

a) Show that; (i) $3y = 1 + 2x^2$ is a solution of the equation $x\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$ (2 marks)

(ii) $-y^2dx + (x^2 + y^2)dy = 0$ is homogeneous (2 marks)

b) Determine the equation of the curve which passes through the points A(1, 1) and B(-1,2) if the gradient at any point on the curve is given by $\frac{d^2y}{dx^2} = 20x^3 + 6x^2$ (5 marks)

c) Solve the equation $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 17y = 0$ (4 marks)

d) Find an integrating factor of the equation $(x^2 + y^2 + x)dx + xydy = 0$ (3 marks)

QUESTION TWO (15 marks):

a) Find the orthogonal trajectory of the family of curves $x^2 = cy^3$ (5 marks)

b) Determine the general solution of each of the following differential equations

(i) $y\frac{dy}{dx} = x^2(1 + y^2)$ (2 marks)

(ii) $2y'' - 7y' + 6y = 0$ (3 marks)

c) A simple pendulum of length ℓ makes an angle θ with the vertical. As it swings back and forth its motion (neglecting friction) is governed by the differential equation

$\frac{d^2\theta}{dt^2} = -\frac{g}{\ell}\sin\theta$ where g (the acceleration due to gravity) is a constant,

Given that $\theta = \theta_0$ and $\frac{d\theta}{dt} = 0$ when t = 0, solve the differential equation, under the assumption that θ is sufficiently small that $\sin\theta$ may be replaced by θ with negligible error. (5 marks)

SECTION B

Answer ANY Three Questions in this Section (39 Marks)

QUESTION THREE (13 Marks)

a) Solve the following differential equation by using the substitution $y = vx$

$$x(x-y)\frac{dy}{dx} + y^2 = 0$$ (5 marks)

b) Given the following differential equation;

$$y'' + 4y' + 4y = 4x^2 + 16x - 2$$

(i) Find the particular solution using the method of undetermined coefficients. (4 marks)

(ii) Find the general solution of the differential equation and find that solution for which $y(0) = 0$ and $y'(0) = 0$. (4 marks)

QUESTION FOUR (13 Marks)

a) Determine $\frac{1}{D^2+3D+2}\left(x^2 + e^{3x}\right)$ and use your result to solve the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = x^2 + e^{3x}$$ (7 marks)

b) A particle is moving in a straight line so that at time t seconds, its displacement x cm from a fixed point O of the line satisfies the differential equation

$$\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 5x = -3e^{2t}$$

When $t = 0$ the particle is at the point where x = -1, moving away from O with speed $2ms^{-1}$

Find x in terms of t and show that $x < 0$ for all $t > 0$ (6 marks)

QUESTION FIVE (13 Marks)

a) Determine if the following differential equation is exact or not.

$$xy^2dx + (x^2y - \cos y)dy = 0$$

Hence or otherwise solve the equation. (4marks)

b) Use the substitution $p = \frac{dy}{dx}$ to reduce the following differential equation to first order and hence solve.

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} = 4x^2$$ (4 marks)

c) Use operator method to find the particular solution of the differential equation

$$(D^2 - 4D + 2)y = e^{2x}(x^4 + x^2)$$ (5 marks)

QUESTION SIX (13 Marks).

a) The differential equation $F(D) = xe^x$ is to be solved using the method of variation of parameters using the following equations where L_1' and L_2' are derivatives of $L_1(x)$ and $L_2(x)$ respectively.

$$L_1'e^{-3x} + L_2'e^{x} = 0 \quad \text{and} \quad -3L_1'e^{-3x} + L_2'e^{x} = xe^x$$

Determine L_1 and L_2 and hence the general solution of the differential equation. (6 marks)

b) Using a suitable change of variable solve the following differential equation

$$\frac{dy}{dx} = \frac{x+1}{x+2y-1}$$ (7marks)

QUESTION SEVEN (13 Marks)

a) At time $t = 0$ the number of bacteria in a certain culture is N_0. At time t the birth-rate of the bacteria is numerically equal to half of the number N of living bacteria and there is a constant death-rate of $\frac{1}{4}N_o$.

i) Assuming N to be a continuous variable, form a differential equation for N and obtain N in terms of t (1mark)

ii) Find the time that elapses before the number of living bacteria doubles. (7marks)

b) Use operator theorems to determine

$$(D^2 - 3D + 2)(e^{3x}\sin 2x)$$ (5 marks)

SECTION A

Answer Both Questions in this Section (31 Marks)

QUESTION ONE (16 marks):

a) Define the following terms relating to a differential equation;

(i) Order – highest ordered derivatives that occurs in a differential equation (1 mark)

(ii) Degree – greated power of highest ordered derivative (1 mark)

(iii) Particular solution (1 mark)

b) Determine the differential equation whose general solution given by;

$y = A + Be^{4t}$ (3 marks)

c) (i) Define a homogenous function f (x, y). (1 mark)

(ii) Determine if the function $f(x,y) = \frac{x^2 - 2xy + 3y^2}{x^3 + xy^2}$ is homogeneous. (2 marks)

(d) Solve the differential equation $y^2 e^x + \frac{dy}{dx} = y^2$ given the initial conditions that $y = 1$ when $x = 0$ (4 marks)

(e) The distance X of a particle from a fixed point P in time t is given by the equation;

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = 0$$

Determine the distance X from P at any time t. (3 marks)

QUESTION TWO (15 marks):

a) For certain values of the constants p, q and r the function, $y = p\sin 2x + q\cos 2x + r$, is a solution of the equation $y'' + 2y' + y = 25\cos 2x + 3$.

Find the values of the constants (4 marks)

b) The rate of increase of the population of a certain town is proportional to that population. If the population of the town was 40,000 in 1980 and was 100,000 in 2016 determine the expected population size in 2030. (6 marks)

c) Show that the equation the following equation is homogeneous and hence solve.

$$2xy\frac{dy}{dx} = x^2 + y^2$$ (5 marks)

SECTION B

Answer ANY Three Questions in this Section (39 Marks)

QUESTION THREE (13 Marks)

a) (i) Prove that the first order differential equation $M(x,y)dx + N(x,y)dy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. (4 marks)

(ii) Determine if the following differential equation is exact or not.

$$(x^2 + 2xy + y)dx + (y^3 + x^2 + x)dy = 0$$

Hence solve the equation. (5 marks)

b) The differential equation $y'' + 2y' - 3y = Q(x)$ is to be solved using the method of undetermined coefficients to find the particular solution y_p

(i) Find the homogeneous solution. (2 marks)

Without solving for y_p write down the general form of y_p if $Q(x)$ is;

(ii) $\sin x$ (1 mark)

(iii) xe^{-3x} (1 mark)

QUESTION FOUR (13 Marks).

a) Determine an integrating factor to the following equation and hence solve it;

$$(x+1)^4 + 2x\,dx - (x+1)\,dy = 0$$ (6 marks)

b) Use the method of undermined coefficients to solve the differential equation;

$$\frac{d^2y}{dx^2} + y = 5\cos x + 2\sin x$$ (7 marks)

QUESTION FIVE (13 Marks)

a) Let F (D) be the function of the linear differential operator associated with the differential equation;

$$ay'' + by' + cy = Q(x) = e^{kx}$$

(i) Show that $F(D) = F(k)e^{kx}$ (2 marks)

(ii) Use (i) to solve $3y'' - 11y' + 6y = e^{4x}$ (4 marks)

b) Reduce the following differential equation to first order and hence solve.

$$(1-x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = x^2$$ (7 marks)

QUESTION SIX (13 Marks).

a) Use a suitable change of variable to transform the following equation to linear form and hence solve it.

$$\frac{dy}{dx} + \frac{1}{x}y = x^2y^3$$ (5 marks)

b) A body of mass m is projected vertically upwards with velocity v_0. If it is subjected to gravity and the air resistance equal to 4 m times its velocity, find height of the body at time t. (8 marks)

QUESTION SEVEN (13 Marks)

a) (i) Define the inverse of the differential operator D. (1 mark)

(ii) Determine $(1+D)^{-1}(x^2+2x+1)$ (3 marks)

b) Solve the following differential equation using the method of the variation of parameters;

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 6x + 3$$ (9 marks)

It state that when completely the plane-polarized light is incident to the analyzer or pass through the analyzer the intensity of the light transmitted by the analyzer is directly proportional to square of the cosine of the angle btwn the transmission axes of the analyzer and the polarizer

Malus law states that when completely the plane polarized light is incident to the analyzer or passes through the analyzer the intensity of the transmitted light by the analyzer is directly proportional to the cosine of the angle between the transmission axes and the polarizer.

SECTION A

Answer Both Questions in this Section (31 Marks)

QUESTION ONE (16 marks):

a) (i) Define an ordinary differential equation (1 mark)

(ii) Determine the order and the degree of the differential equation

$$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^{\frac{3}{5}}-y^{\frac{1}{3}}=0$$ (2 marks)

b) Solve each of the following equations;

(i) $xy\frac{dy}{dx}=x^2+y^2$ (4 marks)

(ii) $2xy^3+3x^2y^2\frac{dy}{dx}=\sin x$ (2 marks)

(iii) $\frac{d^2y}{dx^2}+\frac{dy}{dx}-2y=0$ (3 marks)

c) The tangent of a variable point P on a curve intersects the y-axis at a point Q. For all points P, $\overline{OQ}$ is equal to the value of x at P, where O is the origin. Determine the equation of the family of curves. (4 marks)

QUESTION TWO (15 marks):

a) Determine the differential equation whose general solution given by;

$$y=A+Be^{2x}$$ (3 marks)

b) Show that the following equation is linear and hence solve

$$xdy-ydx=x^2e^x dx$$ (4 marks)

c) Write the following expressions without the D operator

(i) $(D-3)\sin 2x$ (1 mark)

(ii) $\frac{1}{D^2}\left(x^2+e^x\right)$ (1 mark)

d) Solve the differential equation given that $y=0$ and $y'(0)=0$ when $x=0$.

$$y''-6y'+13y=10e^{2x}$$ (6 marks)

SECTION B

Answer ANY Three Questions in this Section (39 Marks)

QUESTION THREE (13 Marks)

a) Determine if the following differential equation is exact or not.

$$(x^3+2xy^2+y)dx+(y^2+2x^2y+x)dy=0$$

Hence solve the equation. (5 marks)

b) The differential equation $y''-3y'-10y=Q(x)$ is to be solved using the method of undetermined coefficients to find the particular solution y_p

(i) Find the homogeneous solution. (2 marks)

Without solving for y_p write down the general form of y_p if $Q(x)$ is;

(ii) x^2+x (1 mark)

(iii) $\cos 3x$ (1 mark)

c) The equation $\frac{d^2s}{dt^2}+100s=0$ represents a simple harmonic motion. Given that $s=10$ and $\frac{ds}{dt}=50$ when $t=0$ determine the distance S as a function of time t.

(4 marks)

QUESTION FOUR (13 Marks).

a) Find an integrating factor to the following equation and hence solve it;

$$3y^2+2xy\frac{dy}{dx}=x+1$$ (6 marks)

b) Use the method of undermined coefficients to find the particular solution of the following equation and hence solve the equation

$$\frac{d^2y}{dx^2}-6\frac{dy}{dx}+9y=4\cos x+2\sin x$$ (7 marks)

$x\frac{dy}{dx} + 2y = y^2x^3$

QUESTION FIVE (13 Marks)

a) Find the particular solution of the equation

$$(D^2 - 4D + 8)y = e^{2x}(x^3 + 1)$$ (5 marks)

b) Use any operator method to solve

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = \sin x$$ (4 marks)

c) Use a suitable transformation to change the following equation to linear form and hence solve it.

$$x\frac{dy}{dx} + 2y = 2xy^2$$ (4 marks)

QUESTION SIX (13 Marks)

a) Use a suitable substitution to transform the following non-homogeneous equation into an equation of separable variables and hence solve.

$$\frac{dy}{dx} = \frac{2x + 2y + 1}{2x - 2y + 1}$$ (5 marks)

b) The rate of change of the temperature of a body is proportional to the difference of the temperature of the body and that of the surrounding air. The initial temperature of the body is $100°C$ and it is reduced to $60°C$ in 1 hour. Assuming that the temperature of the surrounding air to be $40°C$, determine;

(i) the temperature T of the body at any time t (7 marks)

(ii) the temperature of the body at the end of 2 hours (1 mark)

QUESTION SEVEN (13 Marks)

a) (i) Define the inverse of the differential operator D (1 mark)

(ii) Determine $(1-D)^{-1}(x^3 + 4x - 1)$ (3 marks)

b) Solve the following differential equation using the method of the variation of parameters;

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 4x + 2$$ (9 marks)

Instructions: Answer all questions in section A and any OTHER TWO from section B.

SECTION A (30 MARKS)

a) State the three Newton's law of motion [6marks]

b) Find the work done if the point of application of force $\vec{F} = 2\vec{i} + 3\vec{j} - 4\vec{k}$ given in Newton moves point $P(1, 1, -2)$ to $R(4, 2, 5)$ given in metres. [3marks]

c) c) Prove the theorem that the moment of a force about the origin of a coordinate system is equal to the time rate of change of angular momentum. [3marks]

d) A mass of 2kgs is suspended from a spring with a known spring constant of $10N/m$ and allowed to come to rest. It is then set in motion by giving it an initial velocity of $150cm/sec$. Find an expression for the motion of the mass, assuming no air resistance. [4marks]

e) At time $t = 0$ seconds, a rock is dropped from rest from the top of a building 256 feet high. Determine the velocity with which it strikes the ground. [3marks]

f) Due to a force field a particle of mass 5 units moves along a space curve whose position vector is given as a function of time t by $\vec{r} = (2t^3+t)\vec{i}+(3t^4-t^2+8)\vec{j}-12t^2\vec{k}$. Find at any time t the :

i Velocity [2marks]

ii momentum [2marks]

iii acceleration [2marks]

iv force [2marks]

g) Test whether the force field $\vec{F} = x^2yz\vec{i} - xyz^2\vec{k}$ is consevertive or not. [3marks]

SECTION B 40 MARKS

QUESTION TWO 20 marks

a) A particle of mass 2 moves in a field depending on time t given by $\vec{F} = 24t^2\vec{i} + (36t - 16)\vec{j} - 12t\vec{k}$. Assuming that at $t = 0$, $r_0 = 3\vec{i} - \vec{j} + 4\vec{k}$ and has velocity $v_0 = 6\vec{i} + 15\vec{j} - 8\vec{k}$. Determine the:

i torque [5marks]

ii angular momentum [5marks]

b) i) Show that $\vec{F} = (2xy + z^3)\vec{i} + x^2\vec{j} + 3xz^2\vec{k}$ is conservative [4marks]

ii) Find the potential [3marks]

iii) find the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 2)$ [3marks]

QUESTION THREE 20 MARKS

a) A particle is acted on by the forces $\vec{F_1} = 5\vec{i} - 10\vec{j} + 15\vec{k}$, $\vec{F_2} = 10\vec{i} + 25\vec{j} - 20\vec{k}$ and $\vec{F_3} = 15\vec{i} - 20\vec{j} + 10\vec{k}$. Find the force needed to keep the particle in equilibrium [3marks]

b) A particle of mass m moves along a straight line under the influence of a constant force of magnitude $\vec{F}$, if the initial velocity is v_0 find after any time t

i) the speed [3marks]

ii) velocity [2marks]

iii) distance traveled [3marks]

c) A particle is projected from a point on a horizontal plane with a speed 40m/s at an angle θ to the horizontal, where $\tan\theta = \frac{4}{3}$. Find the:

i) Time taken for the particle to return to the plane . [3marks]

ii) Maximum height attained by the particle [2marks]

iii) Horizontal range. [2marks]

iv) Speed of the particle after 2.2 seconds [2marks]

MAT 215

QUESTION FOUR 20 MARKS

a) Aparticle P of mass 2 moves along the x axis attracted toward the origin by a force that is numerically equal to $8x$. If it is initially at rest at $x = 20$, find

i) the differential equation and initial conditions describing the motion [3marks]

ii) the position of the particle at any time t [4marks]

iii) the speed and velocity of the particle at any time $t > 0$ [2marks]

iv) amplitude,period and frequency of the vibration [3marks]

(15)

b) A body of mass 12 kgs is pulled from rest with constant force of 25 Newtons. The force is applied for 6 seconds. Calculate

i) The distance traveled by the body. [3marks]

ii) Final kinetic energy. [2marks]

iii) The final velocity of the body [3marks]

QUESTION FIVE 20 MARKS

a) A particle of mass 5g moves along the x axis under the influence of two forces ; a force of attraction to the origin which in ndymes is numerically equal to 40 times instantaneous distance from the origin O and damping force proportional to the instantaneous speed such that when the speed is 10cm/sec the damping force is 200 dymes. Assuming that the particle start from rest at a distance 20 cm from O

i) Set up the differential equation and initial conditions describing the motion [5marks]

ii) Find the position of the particle at any time $t > 0$ [5marks]

iii) determine the amplitude, period and frequency of the damped oscillation [3marks]

iv) graph the motion [3marks]

b) Find the component of force for the potential function $\phi = 1/2k(x^2 + y^2 + z^2)$ [4marks].

MAT 221 ODE I CAT II

a. Use the method of variation of parameters to find the complete solution of the following differential equations $\frac{d^2y}{dx^2}+y=\sec x$ (5 marks)

b. Use the method of undetermined coefficients to find the complete solutions of the following differential equations $\frac{d^2y}{dx^2}-2\frac{dy}{dx}-3y=xe^{-x}$ (5 marks)

c. Find the general solution of $\frac{d^2y}{dx^2}-2\frac{dy}{dx}-3y=e^{-2x}\cos 3x$ (5marks)

d. Find the particular integral of $(D-1)(D-2)y=\frac{e^{5x}}{D-2}$. 4marks

e. The mass of a vibrating mass is given by $\frac{d^2y}{dx^2}+8\frac{dy}{dx}+20y=300\sin 4t$ .Show that the general solution of the differential equation is given by $y=e^{-4t}(A\cos 2t+B\sin 2t)+\frac{15}{13}(\sin 4t-8\cos 4t)$ 5marks

f. Solve $(D^2-1)y=xe^{3x}$. 3marks

g. Find the particular solution of the differential equation $5\frac{dy}{dx}+2x=3$, given the boundary conditions $y=1\frac{2}{5}$ when $x=2$ 3marks

PHY 212 CAT ONE

1. State the effect on the kinetic energy of the photoelectrons produced with:

(i) Decrease in wavelength of the radiation producing them, (1 mark)

(ii) Increase in the intensity of radiation producing them. (1 mark)

(iii) Making the metal producing the electrons dirtier. (1 mark)

2. Caesium has a work function of 1.9 electronvolts. Find

(i) Its threshold wavelength. (3 marks)

(ii) The maximum energy of the liberated electrons when the metal is illuminated by light of wavelength $4.5 \times 10^{-7} m$. (3 marks)

(iii) The stopping p.d. (3 marks)

(1 electron $= 1.6 \times 10^{-19} J$, $h = 6.6 \times 10^{-34} Js$, $c = 3.0 \times 10^{8} m/s$)

3. state any three properties of X rays. (3 marks)

KARATINA UNIVERSITY
SCHOOL OF PURE AND APPLIED SCIENCES,
DEPT OF MATHEMATICS, STATISTICS & ACTUARIAL SCIENCE
CAT II (MAT 221: ODE I)

Date: Tuesday, 28st March 2023. **Time**: 1600 hrs - 1700 hrs

Instructions: Answer all questions

(1) Use substitution method to solve the special case of homogeneous differential equation given as

$$\frac{dy}{dx} = \frac{x+y+4}{x-y-6}$$

[4 Marks]

(2) Using the method of variation of parameters evaluate

$$y'' - 4y' + 3y = e^{-x}$$

[5 Marks]

(3) Show that

$$y = e^{\alpha x}[A\cos\beta x + B\sin\beta x]$$

whenever $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$ **[3 Marks]**

(4) Solve

$$x^2y'' + 3xy' + 3y = 0$$

[3 Marks]